Ashfaq Mahmud Khan

Prospective PhD Student, Fall 2027 · two phase heat transfer, CFD , explainable machine learning and optimization
Research Portfolio: https://sites.google.com/view/ashfaqmahmudkhan
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Education

Khulna University of Engineering & Technology (KUET), Bangladesh — 2022-Aug 2026
Bachelor of Science (B.Sc.) in Mechanical Engineering
CGPA: 3.36/4.00

Research Interests

Computational Fluid Dynamics • Single and Multiple Phase Heat Transfer • Heat Exchanger Design Optimization • Finite Element Analysis • Data-Driven Modeling & Machine Learning • Battery Thermal Management • Electronics Cooling • Refrigeration & Heat Pump • Complex Fluid-Structure Interaction • Turbulence Control • Renewable Energy Flows • Advanced experimental methods.

Publications

Peer-reviewed Journal Articles

1. **A. M. Khan**, D. Mondal, M. S. Alom, H. Ghosh (2026). "A CFD-based investigation of flow boiling of R32 in helical pipe evaporator: Parametric effects, phase-change behavior, and thermohydraulic benchmarking." *International Communications in Heat and Mass Transfer*, 180, 112826. doi:10.1016/j.icheatmasstransfer.2026.112826
2. Z. Islam, D. Mondal, **A. M. Khan** (2027). "Inclination sensitivity of R32 condensation in smooth tubes: Heat transfer, film redistribution, and pressure gradient." *Next Energy*, 14, 101168. doi:10.1016/j.nxener.2026.101168
3. M. M. M. Niloy, D. Mondal, **A. M. Khan** (2026). "A numerical study with surrogate-assisted multi-objective optimization of cross-flow liquid cooling for cylindrical Li-ion cell array." *Thermal Science and Engineering Progress*, 75, 104774. doi:10.1016/j.tsep.2026.104774
4. A. Das, D. Mondal, **A. M. Khan** (2026). "Numerical investigation of sulfate-induced corrosion in carbon steel pipelines under flow conditions: Effects of velocity, temperature, and flow regimes." *Results in Surfaces and Interfaces*, 24, 100851. doi:10.1016/j.rsurfi.2026.100851
5. R. Sharikul, D. Mondal, T. Nath, **A. M. Khan**, M. A. Islam, M. A. Hasib. (2026). "Evaluation of mechanical and thermal behavior of water hyacinth filled jute/juco composites for aesthetic wall insulation." *Journal of Building Engineering, 134, 117743*. doi:10.1016/j.jobe.2026.117743

Manuscripts Under Review

6. M. R. Islam, D. Mondal, **A. M. Khan**. "Performance evaluation of tapered dual plane twisted-tape inserts with hybrid nanofluids in circular pipes." *Next Energy*.
7. **A. M. Khan**, O. Rahman, S. R. Farabi. "Compressive behavior and failure characteristics of additively manufactured octagonal lattice structures." *Progress in Additive Manufacturing*.
8. O. Rahman, **A. M. Khan**, S. R. Farabi. "Investigation of the deformation mechanisms and energy absorption of 3D-printed PETG Omega structures for lightweight protective applications." *Thin-Walled Structures*.

Manuscript in Preparation

9. **A. M. Khan**, D. Mondal, S. Hasan, T. H. Anto, O. Rahman. "A diverse and explainable machine learning framework for multi-property prediction for R32 in helical pipe evaporators."

Conference Papers

10. T. H. Anto, **A. M. Khan**, S. Shankar, S. Mahmud (2026). "Numerical analysis of thermal hydraulic performance in a staggered tube bank with Al_2O_3–water nanofluid." *3rd International Conference on Mechanical, Manufacturing and Process Engineering (ICMMPE 2026)*, pp. 1–6.
11. M. S. Hossen, D. Mondal, **A. M. Khan**, L. Islam (2025). "Numerical analysis of heat transfer and flow characteristics of R1336mzz(E) with Al_2O_3 nanoparticle in the U-bend tube evaporator." *8th International Conference on Mechanical and Renewable Energy (ICMERE 2025)*.

Research Experience

Undergraduate Research Assistant, KUET (Research Advisor: Dr. Dipayan Mondal) — 2025-2026

Flow boiling of R32 in helical pipe evaporators

- Built 3D volume of fluid simulations with the Lee phase-change model in ANSYS Fluent; validated the heat transfer coefficient against experimental data with a mean deviation of 4.6%.
- Derived a five fold Nusselt correlation from 73 Latin hypercube cases with an R^2 of 0.987.

Explainable machine learning for two-phase flow

- Trained an compared seven machine learning models on 149 CFD cases; XGBoost predicted heat transfer coefficient best (test R^2 = 0.954) and random forest predicted pressure drop best (test R^2 = 0.976).
- Used SHAP, partial dependence and ICE analysis to verify the most influential parameters: mass flux and coil diameter carry 84% of heat transfer attribution.

Dimensionless analysis (Froude number) of R32 condensation

- Wrote a compiled user-defined function (UDF) for interphase mass and energy sources for 11 inclinations (−90° to +90°).

Design optimization of liquid cooling channels for lithium-ion battery modules

- Designed a novel semicircular cross-flow cooling channel for Li-ion cells and optimized it with a response surface surrogate on 42 CFD cases, reducing cell temperature non-uniformity by 69.7%.

Additional

- Pipeline corrosion: CFD with Butler–Volmer kinetics showed temperature controls the rate.
- Bio-based wall insulation: hyacinth filler lowered composite thermal conductivity by 17–24%.
- 3D-printed PETG energy absorbers: Abaqus model matched compressive modulus within 5.2%.

Technical Experience

Thermal-Fluid Laboratory, KUET 2022-2026

- Conducted experiments on heat exchanger, boiling and condensation, refrigeration, heat-pump, radiation and engine test rigs (TecQuipment, Edibon, Plint).

Collaborative CFD Problem-Solving, KUET 2025-2026

- Worked with departmental classmates to diagnose and resolve ANSYS Fluent simulation challenges in their thesis projects, many beyond my own research: biogas combustion, electrochemical corrosion, PCM battery cooling, spoiler airfoils, pulsatile arterial flow and photovoltaic solar radiation.

Private Tutor – 11th, 12th Grade & University Admission

- Tutored 11th–12th grade students in physics, chemistry, and mathematics, and mentored university admission candidates who secured chances at top public universities in the country, using personalized study plans and mock assessments.

Skills

Simulation	ANSYS Fluent (VOF, Lee phase-change model, compiled UDFs, conjugate heat transfer), ANSYS CFX, ANSYS Polyflow, ANSYS optiSLang, ANSYS Mechanical, ICEM CFD Meshing, COMSOL Multiphysics, Abaqus (static implicit)
Data and ML	Python (NumPy, Pandas, scikit-learn, XGBoost, SHAP), Latin hypercube sampling, response surfaces, multi objective optimization.
CAD	SolidWorks (CSWP), AutoCAD.
Programming	Python, C.

Honors, Certifications & Training

- Certified SOLIDWORKS Professional (CSWP) & Associate (CSWA), Dassault Systèmes. 2026
- Workshops in ANSYS Workbench (IMechE KUET) and robotics (LOOP Club). 2023-2026
- Technical Board Scholarship, KUET (merit-based). 2022-2026
- Winner, Project Showcasing Competition, LOOP Club, KUET. 2022

Industrial Attachement & Visit

Bangladesh Industrial Technical Assistance Center (BITAC), Khulna (Industrial Attachment) 2025

- Completed 2 weeks hands-on training in machining, CNC operation, welding and manufacturing processes.

HAMKO Battery, Khulna (Industrial Visit) 2026

- Observed battery manufacturing and assembly lines, including electrode processing, cell assembly and quality testing.

Standardized Test

Test of English as a Foreign Language (TOEFL)

- Scheduled for 18 November 2026.